

PART A**Q1**

a) Business intelligence is the practice of knowing the business, and what to do when based on insights from the daily running of the company, while business analytics is a subdomain of BI, that is more focused on hard facts and measureable units.

b) The difference is that an OLTP system works as a traditional SQL database with inserts, update, read and delete operations to store data in fields, rows and columns and is also normalized. An OLAP system is normally made in order to answer a specific business question and is based on data stored in an OLTP system. An example of an OLTP system would be a telecommunication service storing customer names, customer phone numbers. An example of an OLAP system could be how low prices for subscriptions you could have before not turning a profit, or how high price before you don't get any customers to find out what the "right" price to turn max profits.

c) Descriptive statistics answers what has happened. Predictive statistics predicts what will happen. Prescriptive analytics answers what you should do. An example of a descriptive analytics could be a scatter plot of how often and how much customers buy. An example of predictive statistics could be a linear or logistic regression line chart model which predicts how much revenue one will turn, based on historic data. Prescriptive analytics could be used to answer what one should do, for example if the margin of error is too big in one model it might be better to go for the more conservative estimate.

Q2

a) Intelligence: When you gather the data that should be used

Design: The phase of designing the model or models based on what you need to answer

Choice: The phase of choosing to go for a model

Implementation: To actually physically execute the work that's needed

b) The six phases of the cross industry standard procedure for datamining is 1) gathering the data, 2) staging, 3) ETL, 4) Analyze, 5) Assess, 6) Validate.

The process of gathering the data is to have data to work with. The process of ETL is to uniformize the data if it consists of different types of units, say currency or use different decimal separators. The process of analyzing the data is to get the answer to the business question one is out to solve, and the process of the validation phase is to decide if the model and data can be used practically or if one needs more data, or edit the ETL process and so on, if the data is too uncertain to make decisions off of.

PART B**Q3**

a) Staging, if you have several data sources that you need to manage. Deduplication, if there is many rows that are copies of each other these needs to be deduplicated (uniqueness). Transformation, if there is say several ways to name countries like "America", "US" and "USA", one will need to decide to go for *one* of these.

Q4

a) The standard deviation tells the manager that Product B varies in how much it sells, for the purpose of being clear, it could be that it sells nothing during the winter and sells extremely well in the summer. Product A is more stable as it has a lower standard deviation, and for the sake of being clear, let's say that is indicative of selling as well in the winter as in the summer. (We can't know if this is the case from the text, it's just an example used to explain standard deviation).

b) If it has a long tail to the right, it is left skewing (the top of the curve is to the left). A real world example could be the amount of people globally who live beneath the Norwegian poverty line, or uses less than 30 kr a day globally.

Q5

a)

Completeness, that the dataset has complete rows and not a lot of NULL-values.

Uniqueness, that the dataset does not have duplicate records where it should not (ok if people have same address, not if same PK (unlikely)).

Validity, that the data is valid and represents reality as best as can instead of say an assistant manager INSERT-ing rows which makes the fiscal year results seem good even if they are not just to please the board.

b) For completeness one should have input validation from the get go. If it is not complete, don't insert it until it is. One could also select the incomplete queries, and gather intel to make them complete if they were uncomplete before implementing input validation.

For uniqueness, make a constraint rule in the RDBMS regarding the fields which cannot be

duplicates. For example two different customers should both be able to shop for 20 kr at the same time of day in different point of sales (kasser/shrink(?), but they should not be able to have the same receipt ID.

Validity is perhaps the hardest to enforce as what one measures or registers varies a lot. Say for EAN's one practical action an organization could take is using an infrared scanner to scan the barcode, instead of relying on typing it in, one number by one number. This might swiftly become error-prone. For products without barcodes like loose potatoes, one uses a weight to measure the weight of them to give a price so that one can make sure everyone pays the same amount per kilo of potatoes.

PART C

Q6

a) 1), linear regression. It also shows margin of error. 2) ROC-AUC, it is easy to understand, and by refactoring the order you can use the same model and data to answer both "predicted churn, actual churn", and "predicted no churn, actual no churn".

b) 1). 75% (60 customers actual no churn predicted no churn + 15 predicted churn actual churn out of a 100 customers). The formula is $\text{Cases}/(\text{TP}+\text{TN})$.

2) If there is only $N=100$ it is not a big data pool to make decisions from. The 15 people might be a small company who simply switched to another operator, and we don't know what the timeline of the previous data that the prediction model is based off.

c) To show how good a model is. If the line is 90 degrees right up right down, ($\text{ROC-AUC} = 1.0$) we can see that the model perfectly predicts every true positive and does not get tricked by false positives. If it is 0.5, it is useless, as it would be the same as guessing. It would for example be more important in an Diprivan administering case (giving bedøvelse) that the ROC-AUC is high so that one does not give more than a body can handle and kills them while you just wanted to make them sleep heavy for a while, than it would be for a computer malware application where you in the worst case just could reinstall the operating system, so that is more tolerable to have more false positives, like in the case of key-gens for pirating computer games.

Q7

a) I would use regression, because you use historical data points, where you "regregate" them

into a prediction.

b) I would use classification. Because you compare with other cases you know are fraudulent and not and you would use say cluster classification to compare fraudulent transactions with valid ones as they have certain circumstances and recognizable artefacts.

Q8

a) In reality, data mining mostly consists of numbers and "hard facts", and text mining consists of understanding human communication and meaning (sentiment).

b)

1. The goal of sentiment analysis in that context is to aggregate the most common comments about the product, so if say there's a bad part, the company could make that part better in the next product iteration. Can also be so simple that customers want the product to be available in a color that they don't sell. You would do it this way if there's ten or hundreds of thousands of reviews for a product to systematically go after each bettering point and focusing on that. If the color is less important than the bad part, you focus on fixing the bad part first or you do both in parallel and so on. Sentiment analysis aids in what to do in this regard.

2. What they should manage to improve, and how critical the improvement is (prioritizing).

PART D

Q9

a) Products (A, B) produced per day | Profit per product (A, B) | Time cost relative to profit

b) How to make the most profit the fastest the cheapest. $LP = A + B$.

c) The constraints are how much throughput there is, due to the machines being available only 100 and 80 hours a week instead of 24/7. They don't produce the same products in the same amount of times.

PART E

Q10

a) The difference is the purpose and design. One do uses OLTP for data warehousing (creating fact and dim tables), but data warehouses are commonly denormalized and made to answer business questions by using analytics instead of simply managing data fields. Snowflake schema for deep analytics that won't make do with a star schema, that is much faster and easy to understand for laymen.

b) The fact table is typically a factual measurement, like revenue or amount of sales. Dimension tables will give context to this measurement, and be able to answer amount of sales last year, in total, with another dim table letting one know which seller had the most sale that and that month.

Q11

a) Fact table name: Revenue. It stores the amount of revenue, how much the shop has earned on the bottom line. Dim table 1: Sales, Dim table 2: Seller, Dim table 3: Buyer, Dim table 4: Product.

b) The granularity in a fact table means how fine-grained the data points are. If the grain in the Revenue table in Q11a were coarse, we could only see revenue per year. If the grain were finer, we could see revenue per quarter, month, week, day and hour. So the one example is time grain.

Q12

a) Cleaning. If the data is dirty, it will affect the models, which in turn will make the business take actions based on more uncertain data than necessary. Transforming. This is somewhat related to cleaning, but one need to make sure units are the same. 100NOK and 100USD is very different. 1,000 USD is about 10 000 NOK in the US, but it is only a buck in Norway due to the nature of the decimal separator, even if this data is clean will it make the decision foundation in the future uncertain. The third typical activity is to save "weird" rows into a text file to evaluate later. There might be rows being real and useful, even though they have NULL values, regarding what business questions you are out to answer.

b) An SCD is a field in which is likely to change because people do not live in a vaccum. They might move, they might change names and debit and credit cards. Prices will in endogenic FIAT currency systems always increase due to inflation, and more unlikely go down some times. For name changes and adress changes there might be no point in having and tracking that historical data, but for price changes it is important. SCD Type 1 is overwriting, and SCD

Type 2 is keeping the data.

Q13

a) Velocity, veracity, volume, variety, value. Velocity is how fast data is processed (created, deleted, uploaded, downloaded. Veracity is the "accuracy of re-representing it", e.g. the bits should be in the same place and order as it was before it was uploaded, and after it was downloaded (considering one does not do any compression, this will happen when you upload a YouTube video for example), but say you use Filezilla in a TV corporation to make your news video snippet feature on the news, you want that file to be the same as you exported from the editing software to be the same as is shown in the media presentation controller room. The volume is the vast amounts of data that is exponentially increasing. Variety goes for file types and whether they're unstructured, semistructured or structured data. The value is not as technical as the former, but more for human-centered. Is the data worth it to save? What is it worth to whom? Grandma might put more value in family photos, than what she does in log files from conflicting mods in Skyrim.

b) First, they are two different things. Big data is mainly for processing data (reading and writing data, and storing it). Data warehousing and BI is typically used for (somewhat) time-sensitive analytics and you will replace the data say each month, and let the "old data" be stored in the data warehouse, but you will very unlikely use sales data from the 1970's operations of your business in a daily business operation context. In short you can say that big data is a data source, and DW and BI is refinement.